Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	2 October 2026
Team ID	SWTID-2026-4524
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Team Leader: Swathi R | Team Members: Sandhiya D, Sandhya Charlet Calista S, Mallemkonda Dallyjasmine | Team Size: 4

Technical Architecture:

The architectural diagram of the project and the information as per Table-1 and Table-2 are given below.

Technical Architecture: Implement Client Script & UI Policy (Incident)

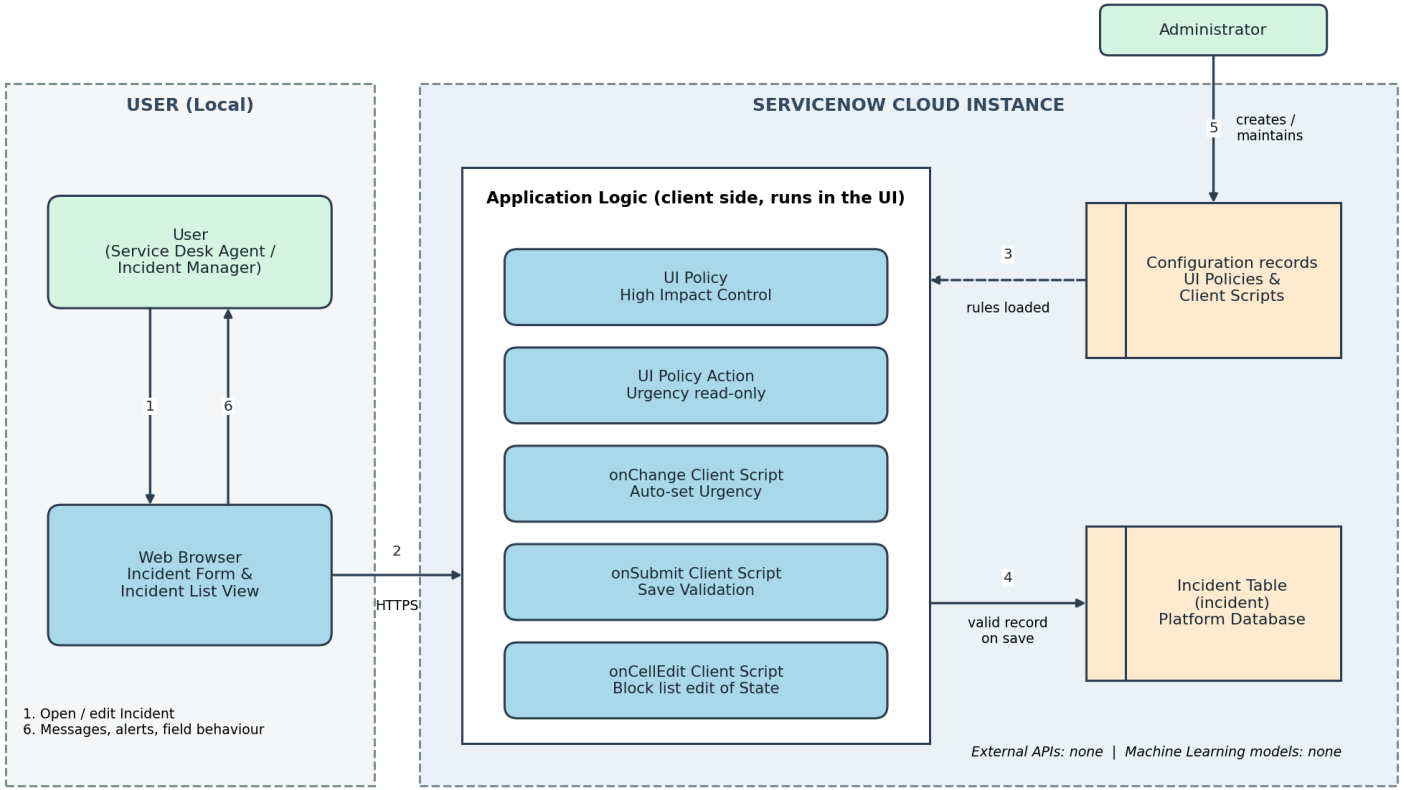


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application: the Incident form and Incident list view opened in a web browser.	ServiceNow Incident form and list (HTML, CSS, JavaScript in the browser)
2.	Application Logic-1	UI Policy "High Impact Control" (Impact is 1 – High) makes Assignment group mandatory, with Reverse if false. Its UI Policy Action makes Urgency read-only.	ServiceNow UI Policy and UI Policy Actions (no-code configuration)
3.	Application Logic-2	onChange script sets Urgency to High when Impact is High. onSubmit script blocks the save when Impact is High and Assigned To is empty.	JavaScript (ServiceNow Client Script, g_form API)
4.	Application Logic-3	onCellEdit script shows an alert and rejects direct edits of the State field in the Incident list view.	JavaScript (ServiceNow Client Script, onCellEdit)
5.	Database	Incident records with the fields Impact, Urgency, Assigned To, Assignment group and State.	ServiceNow Incident table (incident)
6.	Cloud Database	Database service on the cloud, used to store Incident records and the UI Policy and Client Script configuration records.	ServiceNow platform database (managed by the ServiceNow cloud instance)
7.	File Storage	No file storage is needed. The policies and scripts are stored as configuration records in the instance.	Not applicable
8.	External API-1	No external API is used in this project.	Not applicable
9.	External API-2	No external API is used in this project.	Not applicable
10.	Machine Learning Model	No machine learning model is used in this project.	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: only a web browser is needed on the user's computer. Cloud Server Configuration: ServiceNow instance accessed with administrative or configuration access.	ServiceNow cloud instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No third-party open-source framework is added. The client scripts use standard browser JavaScript on the ServiceNow platform.	JavaScript (ECMAScript)
2.	Security Implementations	Only users with administrative or configuration access can create or change UI Policies and Client Scripts. Users log in to the instance, and the read-only Urgency field and blocked list edits protect controlled fields.	ServiceNow roles and access controls (ACL), login authentication, HTTPS
3.	Scalable Architecture	A multi-tier platform (browser, application server, database). The same UI Policy and Client Scripts apply to all Incident records and users, and new rules can be added without redesign.	ServiceNow cloud platform (3-tier)
4.	Availability	The rules run inside the ServiceNow cloud instance, so they are available whenever the Incident form is available. No separate server is needed.	ServiceNow cloud hosting
5.	Performance	The scripts are lightweight and run in the browser, so validation and field updates happen without extra server calls. The onChange script exits early while the form is loading.	Client-side JavaScript (g_form API)